

REACHINBOX / OPERATIONS

Every send, *considered.*

A persistent, rate-aware email scheduler built with BullMQ and Redis.

Product Surface

The dashboard gives authenticated users a focused workspace for scheduling and reviewing individual recipient sends.

EMAIL SCHEDULER / OPERATIONS

Every send, *considered.*

Schedule thoughtful outreach at a pace your audience can respect.

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Welcome back

Sign in to your sending workspace.

Continue with Google →

Configure Google OAuth credentials in the backend environment before signing in.

System Flow

React dashboard

- > Express REST API
- > PostgreSQL + Prisma campaign records
- > Redis + BullMQ delayed jobs
- > Worker: claim, rate-limit, send
- > SMTP provider
- > Elasticsearch index
- > Scheduled / Sent dashboard

Bull Board monitors waiting, active, delayed, completed, and failed jobs.

Implementation Highlights

Persistent scheduling

Delayed jobs live in Redis and survive application restarts.

Idempotency

Atomic database claiming prevents the same EmailJob from being sent twice.

Individual delivery

Each CSV address becomes one EmailJob and one SMTP message.

Controlled throughput

Worker concurrency, inter-email delay, and Redis hourly limits protect the provider.

Local Demo

Run Docker infrastructure, start the API, worker, and Vite frontend, then open **<http://localhost:5173>**. Ethereum provides preview-only testing; Gmail, Brevo, SendGrid, Mailgun, or SES deliver to real inboxes.